

Algebra II Reference and Practice Packet

Context

This packet connects four central Algebra II ideas: quadratic structure, inverse functions, the inverse relationship between exponential and logarithmic functions, and complex numbers. The reference pages summarize the essential rules. The practice test that follows is designed for 45 minutes and contains four multiple-choice questions from each topic.

1. Quadratics

For $f(x) = ax^2 + bx + c$ with $a \neq 0$:

- **Standard form:** $f(x) = ax^2 + bx + c$.
- **Vertex form:** $f(x) = a(x - h)^2 + k$; the vertex is (h, k) and the axis is $x = h$.
- **Factored form:** $f(x) = a(x - r_1)(x - r_2)$; the zeros are r_1, r_2 .
- **Axis and vertex:** $x = -\frac{b}{2a}$ and $y = f\left(-\frac{b}{2a}\right)$.
- **Quadratic formula:** $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$.
- **Discriminant:** $D = b^2 - 4ac$. If $D > 0$, there are two distinct real roots; if $D = 0$, one repeated real root; if $D < 0$, two nonreal conjugate roots.
- **Root relationships:** $r_1 + r_2 = -\frac{b}{a}$ and $r_1 r_2 = \frac{c}{a}$.
- If $a > 0$, the vertex gives a minimum; if $a < 0$, it gives a maximum.

Completing the square uses

$$x^2 + Bx = \left(x + \frac{B}{2}\right)^2 - \left(\frac{B}{2}\right)^2.$$

2. Inverse Functions

An inverse reverses a function's input and output. The notation f^{-1} means inverse function, not reciprocal:

$$f^{-1}(x) \neq \frac{1}{f(x)}.$$

A function has an inverse function on its domain only when it is one-to-one (it passes the horizontal line test).

- To find f^{-1} : write $y = f(x)$, interchange x and y , solve for y , and state any necessary domain restriction.

- Composition check:

$$f(f^{-1}(x)) = x, \quad f^{-1}(f(x)) = x$$

wherever the expressions are defined.

- Domain and range switch:

$$\text{Dom}(f^{-1}) = \text{Range}(f), \quad \text{Range}(f^{-1}) = \text{Dom}(f).$$

- The graphs of f and f^{-1} are reflections across $y = x$.
- A quadratic must have its domain restricted to one side of its vertex before it has an inverse function.

3. Exponential and Natural Logarithmic Functions

The natural exponential function is $\exp(x) = e^x$. Its inverse is $\ln x$:

$$\ln(e^x) = x \quad (x \in \mathbb{R}), \quad e^{\ln x} = x \quad (x > 0).$$

Thus $y = e^x$ and $y = \ln x$ are reflections across $y = x$.

Exponential rules	Logarithm rules
$e^u e^v = e^{u+v}$	$\ln(MN) = \ln M + \ln N$
$\frac{e^u}{e^v} = e^{u-v}$	$\ln\left(\frac{M}{N}\right) = \ln M - \ln N$
$(e^u)^r = e^{ru}$	$\ln(M^r) = r \ln M$

For real logarithms, every logarithm argument must be positive. Also,

$$\log_b x = \frac{\ln x}{\ln b}, \quad b > 0, b \neq 1, x > 0.$$

When solving an exponential equation, isolate the exponential expression and take a logarithm. When solving a logarithmic equation, combine logs when useful, rewrite exponentially, and reject values outside the original domain.

4. Complex Numbers as Pairs of Real Numbers

Let

$$z = a + bi \longleftrightarrow (a, b) \quad \text{and} \quad w = c + di \longleftrightarrow (c, d).$$

These are two ways to write the same complex numbers. On the complex plane, $z = a + bi$ is plotted at (a, b) : a is the real coordinate and b is the imaginary coordinate. In particular,

$$a = a + 0i \longleftrightarrow (a, 0), \quad i = 0 + 1i \longleftrightarrow (0, 1).$$

Thus complex numbers extend the real numbers by adding a second coordinate.

Pair addition is ordinary vector addition. Multiplication needs a special rule because $i^2 = -1$:

$$\begin{aligned} z + w &= (a + c) + (b + d)i && \longleftrightarrow && (a + c, b + d), \\ zw &= (ac - bd) + (ad + bc)i && \longleftrightarrow && (ac - bd, ad + bc). \end{aligned}$$

For example, $i^2 = -1$ becomes $(0, 1)(0, 1) = (-1, 0)$. Complex multiplication is therefore *not* component-by-component vector multiplication.

The conjugate reflects a point across the real axis, and the modulus is its distance from the origin:

$$\bar{z} = a - bi \longleftrightarrow (a, -b), \quad |z| = \sqrt{a^2 + b^2}, \quad z\bar{z} = |z|^2 = a^2 + b^2.$$

For $w = c + di \neq 0$, division is defined by multiplying by $\bar{w} = c - di$:

$$\begin{aligned} \frac{z}{w} &= \frac{(a + bi)(c - di)}{c^2 + d^2} \\ &= \frac{ac + bd}{c^2 + d^2} + \frac{bc - ad}{c^2 + d^2}i \\ &\longleftrightarrow \left(\frac{ac + bd}{c^2 + d^2}, \frac{bc - ad}{c^2 + d^2} \right). \end{aligned}$$

Field vocabulary in Algebra II language

A **field** is a number system in which the familiar rules of arithmetic work: numbers can be added, subtracted, multiplied, and divided by any nonzero number. Each rule below is shown in both $a + bi$ notation and pair notation.

- **Closure:** An operation stays inside the number system. The formulas above show that $z + w$ and zw both have the form $A + Bi \longleftrightarrow (A, B)$, so their real and imaginary parts are still real numbers.
- **Additive identity:** This is the number that changes nothing when added:

$$z + 0 = (a + bi) + (0 + 0i) = a + bi \longleftrightarrow (a, b) + (0, 0) = (a, b).$$

Thus $0 = 0 + 0i \longleftrightarrow (0, 0)$ is the additive identity.

- **Additive inverse:** This is the number that adds to z to make 0:

$$-z = -a - bi \longleftrightarrow (-a, -b),$$

because

$$(a + bi) + (-a - bi) = 0 \longleftrightarrow (a, b) + (-a, -b) = (0, 0).$$

Subtracting a complex number means adding its additive inverse.

- **Multiplicative identity:** This is the number that changes nothing when multiplied:

$$z \cdot 1 = (a + bi)(1 + 0i) = a + bi \longleftrightarrow (a, b)(1, 0) = (a, b).$$

Thus $1 = 1 + 0i \longleftrightarrow (1, 0)$ is the multiplicative identity.

- **Multiplicative inverse:** For $z = a + bi \neq 0$, this is the reciprocal that multiplies with z to make 1:

$$\begin{aligned} z^{-1} &= \frac{1}{z} = \frac{\bar{z}}{z\bar{z}} = \frac{a - bi}{a^2 + b^2} \\ &= \frac{a}{a^2 + b^2} - \frac{b}{a^2 + b^2}i \longleftrightarrow \left(\frac{a}{a^2 + b^2}, \frac{-b}{a^2 + b^2} \right). \end{aligned}$$

Therefore

$$zz^{-1} = 1 \longleftrightarrow (a, b)(a, b)^{-1} = (1, 0).$$

Zero has no multiplicative inverse because division by zero is undefined.

- **Commutative property:** Changing order does not change a sum or product: $z + w = w + z$ and $zw = wz$.
- **Associative property:** Changing grouping does not change a sum or product: $(z + w) + u = z + (w + u)$ and $(zw)u = z(wu)$.
- **Distributive property:** Multiplication distributes across addition: $z(w + u) = zw + zu$.

Because the $a + bi$ operations—equivalently, the pair operations—satisfy all these rules, $\mathbb{C}$ is a field. The pair model adds a visual interpretation without changing the Algebra II arithmetic.

45-Minute Practice Test

Instructions. Complete all 16 questions in 45 minutes. Select the best answer for each question. Show supporting work on separate paper. Use exact values unless a question requests an approximation. Unless your teacher says otherwise, complete the test without referring to the preceding reference pages.

Quadratics

1. A quadratic has zeros -2 and 5 and satisfies $f(1) = -24$. What is its minimum value?
A. -49
B. $-\frac{49}{2}$
C. -24
D. $\frac{3}{2}$
2. The equation $x^2 - (k + 2)x + 2k = 0$ has exactly one real solution. What is the sum of k and that solution?
A. 2
B. 3
C. 4
D. 6
3. The graph of $f(x) = ax^2 + bx + 6$ has axis of symmetry $x = 2$ and vertex value $f(2) = -10$. What is the larger zero of f ?
A. $2 - \frac{\sqrt{10}}{2}$
B. $2 + \frac{\sqrt{10}}{2}$
C. $4 - \sqrt{10}$
D. $4 + \sqrt{10}$
4. A projectile's height in feet is $h(t) = -16t^2 + vt + 6$. If its height after 1 second is 70 feet, what is its maximum height?
A. 86 ft
B. 96 ft
C. 106 ft
D. 112 ft

Inverse Functions

5. If $f(x) = \frac{3x - 5}{2x + 1}$, what is $f^{-1}(4)$?
A. $-\frac{9}{5}$
B. $-\frac{5}{9}$
C. $\frac{5}{9}$

D. $\frac{9}{5}$

6. The function $f(x) = (x - 4)^2 - 7$ is restricted to $x \geq 4$. Which expression gives its inverse, including the correct inverse domain?
- A. $f^{-1}(x) = 4 + \sqrt{x + 7}, x \geq -7$
 - B. $f^{-1}(x) = 4 - \sqrt{x + 7}, x \geq -7$
 - C. $f^{-1}(x) = \sqrt{x + 7} - 4, x \geq -7$
 - D. $f^{-1}(x) = 4 + \sqrt{x - 7}, x \geq 7$
7. A one-to-one function satisfies $f(3) = 11$. If $g(x) = f^{-1}(2x - 1)$, what is $g(6)$?
- A. 3
 - B. 5
 - C. 6
 - D. 11
8. If $f(x) = 2(x - 1)^3 + 5$, which expression equals $f^{-1}(x)$?
- A. $1 + \sqrt[3]{\frac{x - 5}{2}}$
 - B. $1 + \frac{\sqrt[3]{x - 5}}{2}$
 - C. $\sqrt[3]{\frac{x - 1}{2}} + 5$
 - D. $\frac{\sqrt[3]{x - 5} + 1}{2}$

Exponential and Logarithmic Functions

9. What is the solution of $\ln(x - 1) + \ln(x + 3) = \ln 18$?
- A. -7
 - B. -3
 - C. 1
 - D. 3
10. Which set contains all real solutions of $e^{2x} - 5e^x + 6 = 0$?
- A. $\{2, 3\}$
 - B. $\{\ln 2, \ln 3\}$
 - C. $\{-\ln 2, -\ln 3\}$
 - D. $\{\ln 6\}$
11. An exponential function $q(t) = ab^t$ satisfies $q(1) = 12$ and $q(4) = 324$. What is $q(0)$?
- A. 2
 - B. 3
 - C. 4
 - D. 9
12. What is the exact solution of $2^{x+1} = 5^{2x-1}$?
- A. $\frac{\ln 10}{\ln(25/2)}$

- B. $\frac{\ln(5/2)}{\ln 10}$
- C. $\frac{\ln(10/25)}{\ln 10}$
- D. $\frac{\ln(25/2)}{\ln 10}$

Complex Numbers

13. On the complex plane, z is represented by $(3, -2)$ and w by $(1, 4)$. Which point represents zw ?
- A. $(3, -8)$
 - B. $(5, 2)$
 - C. $(11, 10)$
 - D. $(11, -10)$
14. Which ordered pair represents $\frac{4 + 2i}{1 - i}$?
- A. $(1, 3)$
 - B. $(1, -3)$
 - C. $(3, 1)$
 - D. $(3, -1)$
15. If $z = -3 + 4i$, which expression equals z^{-1} ?
- A. $\frac{-3 + 4i}{5}$
 - B. $\frac{-3 - 4i}{5}$
 - C. $\frac{-3 + 4i}{25}$
 - D. $\frac{-3 - 4i}{25}$
16. Let $z = x + yi$. If $(2 - i)z + \bar{z} = 7 + i$, what is z ?
- A. $\frac{3}{2} + \frac{5}{2}i$
 - B. $\frac{5}{2} + \frac{3}{2}i$
 - C. $2 + 3i$
 - D. $3 + 2i$